External ear & tympanic membrane

Auricle, external canal and the drum: cartilage, skin, nerves, lymphatics and the otoscopic landmarks you check every day.

1. Auricle
2. Nerve supply of the auricle and canal
3. External auditory canal
4. Lymphatics of the external ear
5. Tympanic membrane
6. In theatre
7. On imaging

01 Auricle

- Single piece of **yellow elastic cartilage** covered with skin. The **lobule has no cartilage** (fat and fibrous tissue only).
- Skin is **tightly adherent on the lateral surface**, looser on the medial surface. Keratinising squamous epithelium with sebaceous glands and hair follicles.
- **Incisura terminalis:** the gap between the tragus and crus of the helix; it has no cartilage, so an endaural incision here avoids cutting cartilage.

CLINICAL PEARL Because lateral skin is stuck to the perichondrium, a haematoma lifts the perichondrium off the cartilage and cuts its blood supply. Drain early to avoid cauliflower ear.

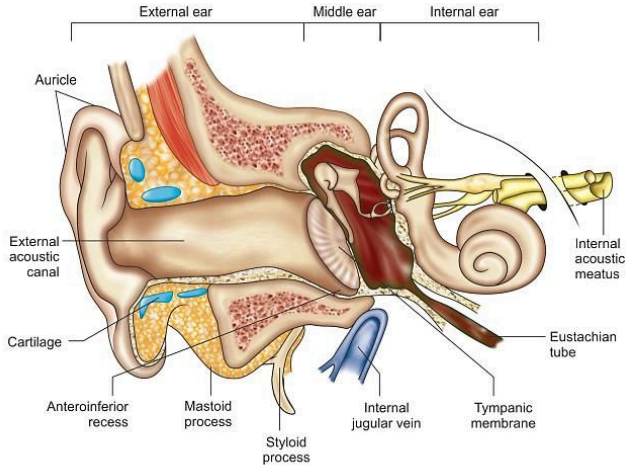
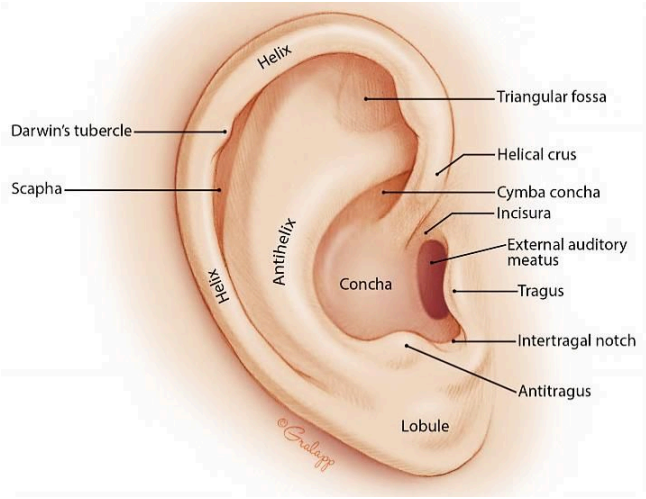


Fig. 6: Three parts of the ear: external, middle and internal

Three parts of the ear: external, middle and internal
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Auricle: surface landmarks
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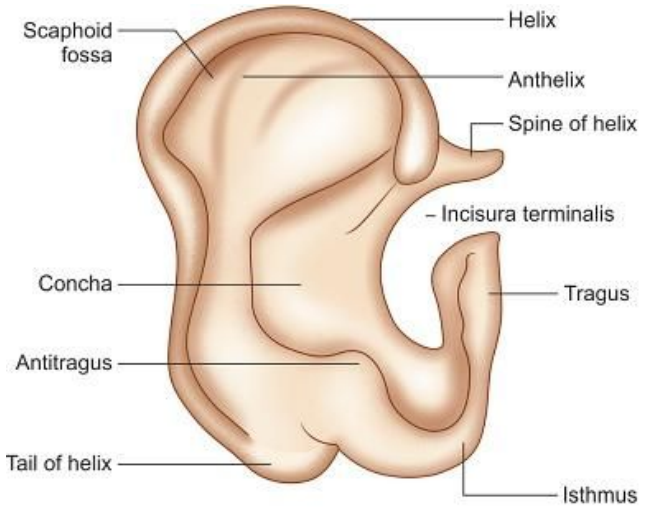


Fig. 8: Auricle cartilage: external features

Auricle cartilage: external features

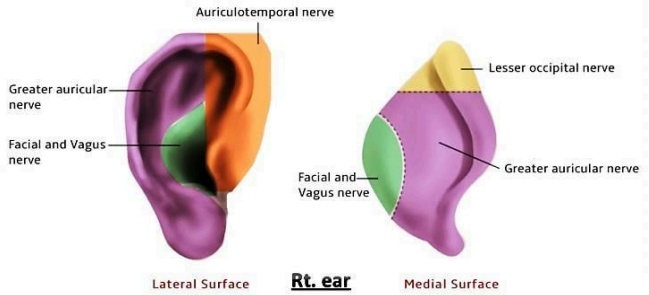
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02 Nerve supply of the auricle and canal

- **Auriculotemporal (V3):** front of the lateral surface (tragus, crus of helix) and the anterosuperior canal wall; anterior half of the lateral TM.
- **Great auricular (C2, C3):** most of the medial surface and the back of the lateral surface, lobule.
- **Lesser occipital (C2):** upper part of the medial surface.
- **Arnold's nerve (auricular branch of X):** concha, posteroinferior canal wall and posterior half of the lateral TM.
- **Facial nerve (VII):** sensory twigs to the concha and posterior canal wall.
- **Jacobson's nerve (IX)** supplies the medial (mucosal) surface of the TM.

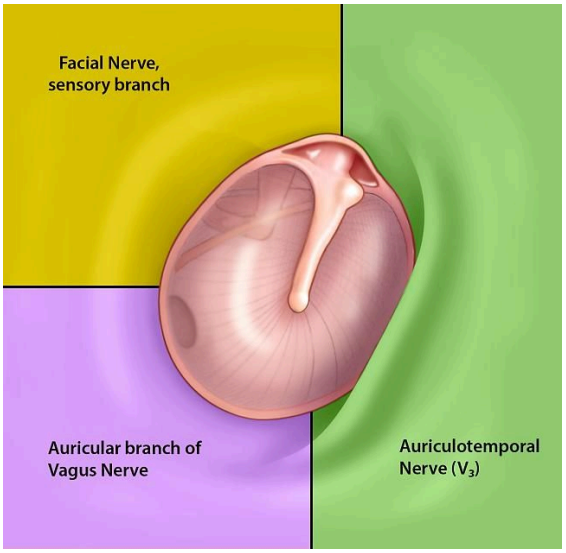
CLINICAL PEARL Hitselberger's sign: numbness of the posterior canal wall in acoustic neuroma (VII sensory fibres affected early). Cleaning the canal can trigger the vagal cough or fainting reflex through Arnold's nerve.

SOURCES DIFFER AURICLE: ENT Steps uses a regional pattern (as here). Doss uses halves: auriculotemporal upper outer half, lesser occipital upper inner half, great auricular lower half of both surfaces.



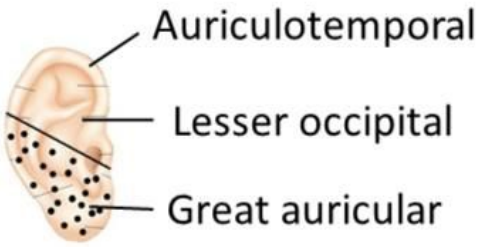
Nerve supply of the auricle

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Sensory supply of the EAC and tympanic membrane

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Sensory supply of the auricle

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03 External auditory canal

- **24 mm long, S-shaped.** The anterior wall is about 6 mm longer than the posterior because the TM is oblique.
- **Cartilaginous outer third (8 mm):** runs up, back and in. Thick skin with **hair, ceruminous and pilosebaceous glands** (wax and furuncles occur only here).
- **Bony inner two-thirds (16 mm):** runs down, forward and in. Thin skin, no hair or glands, continuous with the TM.
- Straighten the adult canal by pulling the pinna **up and back**; in infants, **down and back**.
- **Isthmus:** narrowest part of the bony canal, about 6 mm lateral to the TM. The **anterior recess** lies medial to it and traps debris.
- **Fissures of Santorini** in the cartilage and the **foramen of Huschke** (anteroinferior bony defect, mainly in children) allow spread between the canal and the parotid/TMJ.
- **Relations:** above, middle cranial fossa; below, parotid; front, TMJ; behind, mastoid air cells and facial nerve.
- Skin migrates **from medial to lateral**, which makes the canal self-cleaning.

CLINICAL PEARL Malignant otitis externa spreads through the fissures of Santorini to the skull base, and the first cranial nerve affected is usually VII at the stylomastoid foramen.

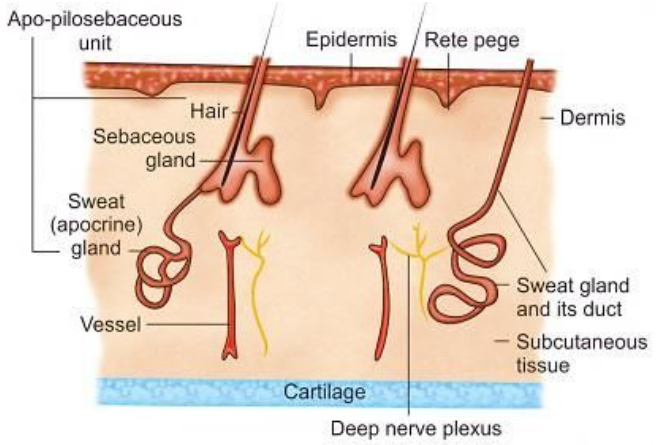
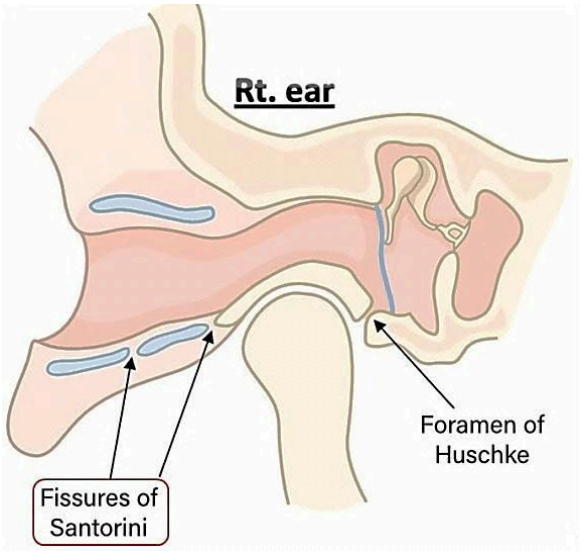


Fig. 10: Skin of cartilaginous external auditory canal

Skin of the cartilaginous EAC with ceruminous and sebaceous glands

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EAC: fissures of Santorini and foramen of Huschke

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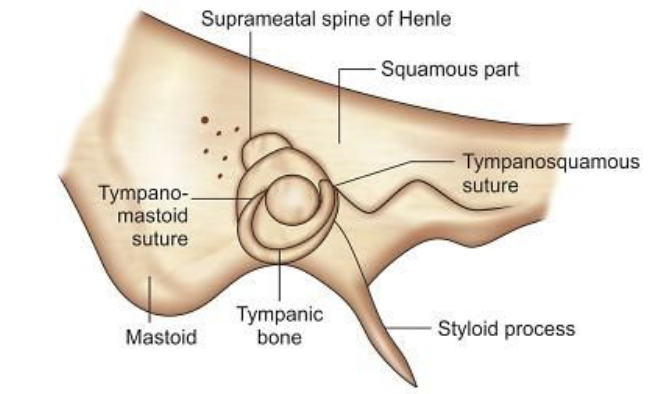
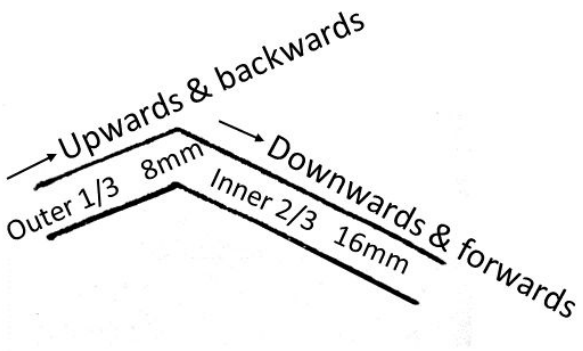


Fig. 11: Lateral view of temporal bone showing endomeatal spines and sutures

Lateral temporal bone: spine of Henle and EAC sutures

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Direction of the external auditory meatus

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04 Lymphatics of the external ear

- Tragus, front of the auricle and anterior canal → **preauricular (parotid)** nodes.
- Back of the auricle → **postauricular (mastoid)** nodes.
- Lobule and floor of the canal → **infra-auricular** nodes, then upper deep cervical nodes.
- Middle ear and ET drain to the **retropharyngeal** nodes. The **inner ear has no lymphatics**.

05 Tympanic membrane

- **9–10 mm high, 8–9 mm wide, 0.1 mm thick.** Set obliquely at **55°** to the canal floor; the posterosuperior part is more lateral.
- **Pars tensa:** most of the drum, with a thickened rim, the **annulus**, sitting in the tympanic sulcus. The **umbo** is the tip of the malleus handle; the **cone of light** points anteroinferiorly.
- **Pars flaccida (Shrapnell's):** above the lateral process of the malleus, between the anterior and posterior malleal folds and the **notch of Rivinus** (no annulus here). Slightly pink.

- **Three layers:** outer epithelium (from canal skin), middle fibrous layer (radial, circular and parabolic fibres; poorly organised in the pars flaccida), inner mucosa (from the middle ear).
- Seen through a normal drum: long process of incus, incudostapedial joint, round window niche and the ET orifice.
- **Mobility** is tested with Siegel's pneumatic speculum.

> **CLINICAL PEARL** Pars flaccida lacks an organised fibrous layer, so attic retraction pockets and cholesteatoma begin here.

> **SOURCES DIFFER PARS FLACCIDA:** ENT Steps says its fibrous layer is very thin and unorganised; Doss says it is absent. **SIZE:** ENT Steps 9–10 × 8–9 mm; Doss 10 × 8 mm.

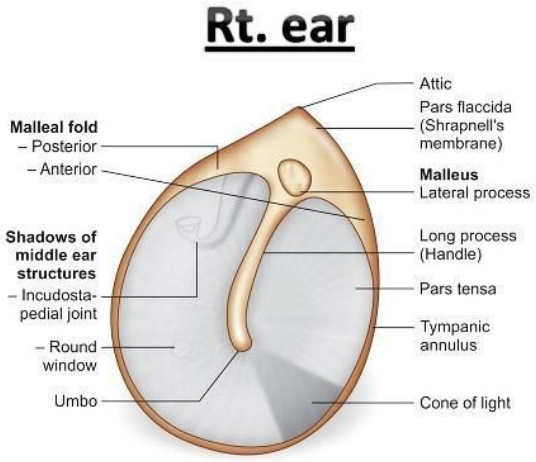
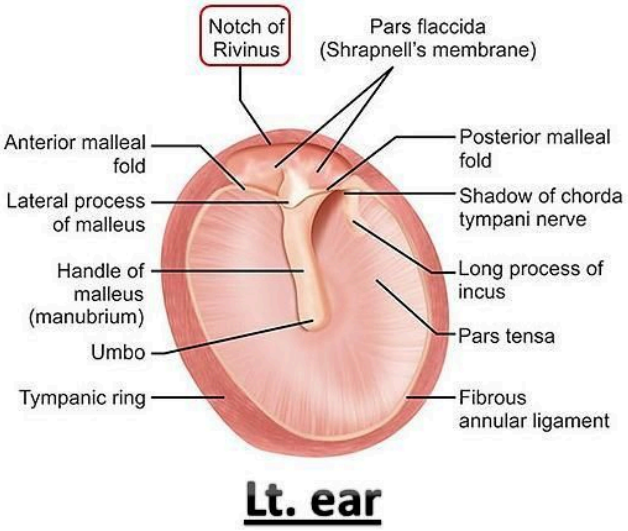


Fig. 13: Tympanic membrane showing attic, malleus handle, umbo, cone of light and structures of middle ear seen through it on otoscopy

Tympanic membrane and middle ear structures seen through it

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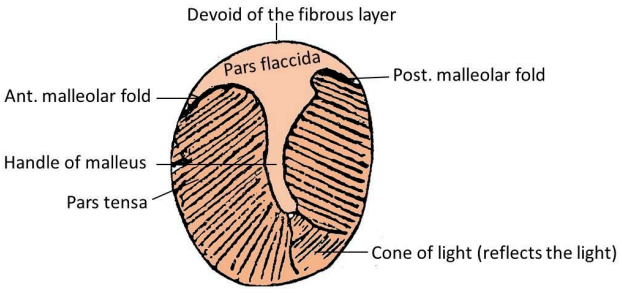
Parts of the tympanic membrane

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Otoscopic view of a normal left tympanic membrane

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Tympanic membrane, lateral view

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06 Surgical anatomy: In theatre

- **Incisions:** endaural through the incisura terminalis (no cartilage), postaural about 1 cm behind the sulcus, or transcanal.

- **Tympanomeatal flap** is raised from the bony canal down to the annulus; enter the middle ear posteriorly where the annulus is thickest, avoiding the chorda tympani.
- **Canaloplasty** widens the bony canal; protect the TMJ in front and the facial nerve and mastoid behind.
- **Graft materials:** temporalis fascia, tragal or conchal cartilage and perichondrium.

> CLINICAL PEARL In children, the bony canal is short and the TM more horizontal, so instruments reach the middle ear sooner than expected.

07 Radiological anatomy: On imaging

- CT shows canal stenosis, exostoses (broad-based, bilateral, cold water), osteomas (single, pedunculated) and keratosis obturans or canal cholesteatoma (bone erosion).
- In **malignant otitis externa**, CT shows bone erosion at the skull base; MRI shows soft tissue spread; technetium and gallium scans show activity and response.

T Tables

Sensory supply of the ear

Area	Nerve	Origin
Tragus, front of auricle, anterosuperior EAC, anterior lateral TM	Auriculotemporal	V3
Concha, posteroinferior EAC, posterior lateral TM	Arnold's (auricular branch)	X
Concha and posterior canal wall (small area)	Sensory twigs of the facial nerve	VII
Most of the medial auricle, lobule, back of lateral surface	Great auricular	C2, C3
Upper medial auricle and postauricular skin	Lesser occipital	C2
Medial surface of TM, middle ear, mastoid, bony ET	Jacobson's (tympanic plexus)	IX

Referred otalgia comes through V3, VII, IX, X and C2–C3. Throat cancer reaches the ear through IX and X.

M Mnemonics

Nerves of the auricle
GALA
Great auricular, Auriculotemporal, Lesser occipital, Arnold's

EAC proportions
One-third cartilage, two-thirds bone
8 mm cartilaginous and 16 mm bony, 24 mm in all

TM layers
Skin, fibre, mucosa (outside in)
Outer epithelium, middle fibrous, inner mucosal

Q Rapid recall

1. **Length of the EAC and the proportion that is cartilaginous?** 24 mm; outer third (8 mm) cartilaginous, inner two-thirds (16 mm) bony.
2. **Where are wax glands and hair found in the EAC?** Only in the cartilaginous outer third.
3. **What are the fissures of Santorini and the foramen of Huschke?** Fissures in the cartilaginous canal floor, and a developmental defect in the anteroinferior bony canal; both let infection or tumour pass to the parotid/TMJ region.

4. **What is Hitselberger's sign?** Hypoaesthesia of the posterior meatal wall from pressure on facial nerve sensory fibres, as in acoustic neuroma.
5. **Which nerve causes the cough reflex when cleaning the ear?** Arnold's nerve, the auricular branch of the vagus.
6. **Dimensions and angle of the tympanic membrane?** 9–10 mm high, 8–9 mm wide, 0.1 mm thick; 55° to the canal floor.
7. **Boundaries of the pars flaccida?** Above the lateral process of malleus, between the anterior and posterior malleal folds and the notch of Rivinus.
8. **Three layers of the tympanic membrane?** Outer epithelium, middle fibrous layer (radial, circular, parabolic fibres), inner mucosa.
9. **What is the incisura terminalis?** The cartilage-free gap between the tragus and crus of the helix, used for endaural incisions.
10. **Direction of the cartilaginous and bony EAC?** Cartilaginous: up, back and in. Bony: down, forward and in.
11. **How do you straighten the canal in adults and infants?** Adults: pull the pinna up and back. Infants: down and back.
12. **What is the isthmus of the EAC?** The narrowest part of the bony canal, about 6 mm lateral to the TM.
13. **What is the anterior recess of the EAC?** A dip in the anteroinferior bony canal medial to the isthmus where debris and foreign bodies collect.
14. **Lymph drainage of the tragus and front of auricle?** Preauricular (parotid) nodes.
15. **Lymph drainage of the lobule?** Infra-auricular nodes, then the upper deep cervical chain.
16. **Which part of the TM lacks an annulus?** The pars flaccida (at the notch of Rivinus).
17. **What is the umbo?** The point of maximal inward tenting of the TM at the tip of the malleus handle.
18. **Which middle ear structures can be seen through a normal TM?** Long process of incus, incudostapedial joint, round window niche, ET opening, sometimes the chorda tympani.
19. **Direction of skin migration in the EAC?** From the TM outward (medial to lateral), making the canal self-cleaning.
20. **What is the pH of cerumen?** Slightly acidic, about 6.0–6.5.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.